

Osman Şahin Güler

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EDUCATION

- **Yaşar University** | September 2024 – Present
- **B.Sc. in Software Engineering** | **GPA: 3.61** | 2nd Year Student

EXPERIENCE & PROJECTS **Yaşar-Anafrata UAV Team** *Autonomous Systems & Software Developer* | Izmir, Turkey

- **Computer Vision Applications:** Developed object detection pipelines using Pinhole Camera Models and YOLO architectures specifically optimized for small object detection from high altitudes.
- **Autonomous Mission Management:** Established telemetry data processing systems via the ArduPilot ecosystem and MAVLink protocol for autonomous navigation and target-oriented UAV missions.
- **Simulation-Based Development:** Performed real-time physical simulations and mission tests in Gazebo and SITL (Software In The Loop) environments to validate flight algorithms.
- **Embedded System Optimization:** Applied hardware acceleration and low-latency data streaming techniques using GStreamer to enhance image processing performance on Raspberry Pi and other edge devices.
- **Modern Infrastructure & Containerization:** Utilized Docker-based microservice architectures and Portainer management systems to ensure environmental consistency during development and deployment.

CLB Automation *Software Developer (Intern/Prototype)* | January 2025 – October 2025

- Developed computer vision prototypes focused on object detection using OpenCV and YOLO.
- Optimized data processing pipelines to increase real-time detection accuracy.
- Conducted feasibility studies by performing model training and testing on custom datasets.
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TECHNICAL SKILLS

- **Programming Languages:** Python (Proficient), C++, JavaScript (React), Java, C
- **AI & Computer Vision:** OpenCV, YOLO (v5, v8, v11), SAHI, NumPy
- **Control & Simulation:** ArduPilot, Gazebo, MAVLink, Mission Planner
- **Tools & DevOps:** Docker, Portainer, Git, Linux (Ubuntu), Raspberry Pi
- **Engineering:** Data Structures & Algorithms, Object-Oriented Programming (OOP)

LANGUAGES & INTERESTS

- **Languages:** English (Fluent), Japanese (Beginner)
 - **Hobbies:** Sailing (Professional athlete for 10 years), Martial Arts
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